Projection and Matrix Inversion

1 Projection

1.1 Orthogonal complement of a subspace

Definition 1 *Let $U \subseteq \mathbb{R}^n$. Then a set $U^\perp$ is said to be the **orthogonal complement** of U if*

$$U^\perp = \{x \ : \ x^T y = 0 \ \forall \ y \in U\}.$$

Note that $U^\perp$ is a subspace.

Proposition 1 *If $\mathbf{A}$ is a $n \times k$ matrix of rank $r < k$, then $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^T)$, and $dim(\mathcal{C}(\mathbf{A})^\perp) = n - r$.*

Proof: Notice that:

$$z \in \mathcal{C}(\mathbf{A})^\perp \Leftrightarrow z^T \mathbf{A} c = 0 \ \forall c \in \mathcal{C}(\mathbf{A}) \Leftrightarrow z^T \mathbf{A} = 0 \Leftrightarrow z \in \mathcal{N}(\mathbf{A}^T).$$

Proposition 2 *For any real matrix $\mathbf{X}$, $\rho(\mathbf{X}) = \rho(\mathbf{X}\mathbf{X}^T) = \rho(\mathbf{X}^T\mathbf{X})$.*

Proof: Note that $\rho(\mathbf{X}\mathbf{X}^T) = \rho(\mathbf{X}) - dim(\mathcal{C}(\mathbf{X}^T) \cap \mathcal{N}(\mathbf{X}))$. We show that $dim(\mathcal{C}(\mathbf{X}^T) \cap \mathcal{N}(\mathbf{X})) = 0$. Suppose $c \in \mathcal{C}(\mathbf{X}^T) \cap \mathcal{N}(\mathbf{X})$. Then $c = \mathbf{X}^T d$ for some d and $\mathbf{X}c = 0$. Now $\mathbf{X}\mathbf{X}^T d = 0$, so $d^T \mathbf{X}\mathbf{X}^T d = 0$. Hence $c^T c = 0$, which implies that $c = 0$.

1.2 Linear projection

Definition 2 *Let U_1 and U_2 be subspaces of vector space V. The **sum** of these subspaces, denoted by $U_1 + U_2$ is defined as:*

$$U_1 + U_2 = \{u_1 + u_2 \ : \ u_1 \in U_1, u_2 \in U_2\}.$$

Note that $U_1 + U_2$ is a subspace of V. The sum of two subspaces U_1 and U_2 is said to be **direct** if any vector in $U_1 + U_2$ can be expressed in a unique way as $u_1 + u_2$ with $u_1 \in U_1$ and $u_2 \in U_2$. When the sum of U_1 and U_2 is direct, we denote $U_1 + U_2$ by $U_1 \oplus U_2$. Note that, the sum $U_1 + U_2$ is direct if $U_1 \cap U_2 = \{0\}$.

Example 1 *$U_1 = \{(x, 0) \ : \ x \in \mathbb{R}\}$ and $U_2 = \{(y, y) \ : \ y \in \mathbb{R}\}$ are subspaces of $\mathbb{R}^2$. Then $U_1 + U_2 = \mathbb{R}^2$, since given any vector $x \in \mathbb{R}^2$, one can write, $x = (x_1, x_2)^T = (x_1 - x_2)(1, 0)^T + x_2(1, 1)^T = u_1 + u_2$. Clearly, $u_1 \in U_1$ and $u_2 \in U_2$. Moreover, $(1, 0)^T$ and $(1, 1)^T$ together form a basis of $\mathbb{R}^2$. So the representation of x above is unique.*

Definition 3 *The **projection operator** that projects the subspace $W := U_1 \oplus U_2$ into U_1 parallel to U_2 is the linear transformation p:*

$$p \ : \ (u_1 + u_2) \to \ p(u_1 + u_2) = u_1, \qquad u_1 \in U_1, \ u_2 \in U_2.$$

p is a linear transformation. If W is a subspace of $\mathbb{R}^n$, for a fixed basis of $\mathbb{R}^n$, we can express the projection operator p in terms of an $n \times n$ matrix $\mathbf{P}$ determining the linear transformation p. Then $\mathbf{P}$ is referred to as the projection matrix corresponding to p.

Proposition 3 *The following statements about an $n \times n$ matrix $\mathbf{P}$ are equivalent.*

1. $\mathbf{P}$ *is a projection matrix.*
2. $\mathbf{P}$ *is idempotent, i.e.,* $\mathbf{P}^2 = \mathbf{P}$.
3. $\mathcal{N}(\mathbf{P}) = \mathcal{C}(\mathbf{I} - \mathbf{P})$.
4. $\rho(\mathbf{P}) + \rho(\mathbf{I} - \mathbf{P}) = n$.
5. $\mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{I} - \mathbf{P})$ *is a direct sum of subspaces.*

Proof:

(1) $\Rightarrow$ (2) Let $\mathbf{P}$ be the projection operator into U_1 along U_2. Then for every $x \in \mathbb{R}^n$, $\mathbf{P}x \in U_1$, so $\mathbf{PP}x = \mathbf{P}x$. Hence $\mathbf{P}^2 = \mathbf{P}$.

(2) $\Rightarrow$ (3) By (2), $\mathbf{P}(\mathbf{I} - \mathbf{P}) = 0$, so $\mathcal{C}(\mathbf{I} - \mathbf{P}) \subseteq \mathcal{N}(\mathbf{P})$. However, if $x \in \mathcal{N}(\mathbf{P})$ then $\mathbf{P}x = 0$, so $x = (\mathbf{I} - \mathbf{P})x \in \mathcal{C}(\mathbf{I} - \mathbf{P})$. Thus $\mathcal{N}(\mathbf{P}) \subseteq \mathcal{C}(\mathbf{I} - \mathbf{P})$ and the equality follows.

(3) $\Rightarrow$ (4) Clearly, $\nu(\mathbf{P}) = \rho(\mathbf{I} - \mathbf{P}) = n - \rho(\mathbf{P})$.

(4) $\Rightarrow$ (5) $n = \rho(\mathbf{I}) = \rho(\mathbf{P} + \mathbf{I} - \mathbf{P}) = \rho(\mathbf{I} - \mathbf{P}) + \rho(\mathbf{P})$. So $\mathcal{C}(\mathbf{P}) \cap \mathcal{C}(\mathbf{I} - \mathbf{P}) = \{0\}$. So the sum is direct.

(5) $\Rightarrow$ (1) Since any vector $x \in \mathbb{R}^n$ can be written as $\mathbf{P}x + (\mathbf{I} - \mathbf{P})x$, so $\mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{I} - \mathbf{P}) = \mathbb{R}^n$. So, by hypothesis, $\mathcal{C}(\mathbf{I} - \mathbf{P})$ is a complement of $\mathcal{C}(\mathbf{P})$ and $\mathbf{P}x$ is the projection into $\mathcal{C}(\mathbf{P})$ along $\mathcal{C}(\mathbf{I} - \mathbf{P})$.

1.3 Orthogonal projection

Definition 4 *If S is a subspace of $\mathbb{R}^n$ and $x \in \mathbb{R}^n$, the projection of x into S along $S^\perp$ is called the* ***orthogonal projection*** *of x into S. $\mathbf{P}$ is an orthogonal projection operator (or, orthogonal projector) if it is the orthogonal projection into some subspace S of $\mathbb{R}^n$.*

It is clear that $S = \mathcal{C}(\mathbf{P})$ and $S^\perp = \mathcal{C}(\mathbf{I} - \mathbf{P})$.

Proposition 4 *Let $\mathbf{P}$ is a $n \times n$ matrix. Then the following are equivalent:*

1. $\mathbf{P}$ *is an orthogonal projector.*
2. $\mathbf{P^T P} = \mathbf{P}$
3. $\mathbf{P}^T = \mathbf{P}$ *and* $\mathbf{P}^2 = \mathbf{P}$

Proof: To show (1) $\Leftrightarrow$ (2), note that $\mathbf{P}$ is an orthogonal projector if and only if $(\mathbf{I} - \mathbf{P})x \perp \mathcal{C}(\mathbf{P})$ for all $x \in \mathbb{R}^n$. This is true if and only if $y^T \mathbf{P}^T (\mathbf{I} - \mathbf{P})x = 0$ for all x and y in $\mathbb{R}^n$. This is true if and only if $\mathbf{P}^T(\mathbf{I} - \mathbf{P}) = 0$. Equivalence of (2) and (3) is trivial.

Proposition 5 *Let $\mathbf{P}$ be a orthogonal projector on a k-dimensional subspace S of $\mathbb{R}^n$ ($k \leq n$), and suppose that x_1, x_2, …, x_k form a basis for S. Then*

$$\mathbf{P} = \mathbf{X}(\mathbf{X^T X})^{\mathbf{-1}}\mathbf{X^T},$$

where $\mathbf{X} = [x_1 : x_2 : \cdots : x_k]$ is an ($n \times k$ matrix).

Proof: Consider the projection $\mathbf{P}$ applied to any $y \in \mathbb{R}^n$. Since $\mathbf{P}y \in S$ there exist coefficients β_1, β_2, …, β_k such that:

$$\mathbf{P}y = \beta_1 x_1 + \cdots + \beta_k x_k = \mathbf{X}\beta.$$

Further $y = \mathbf{P}y + (\mathbf{I} - \mathbf{P})y$, and $(\mathbf{I} - \mathbf{P})y \in S^{\perp}$. Thus each of $x - 1$, x_2, …, x_k is orthogonal to $(\mathbf{I} - \mathbf{P})y$. Thus $\mathbf{X}^T(\mathbf{I} - \mathbf{P})y = 0$. So it follows that $\mathbf{X}^T(\mathbf{y} - \mathbf{X}\beta) = 0$. Solving for β, we get $\beta = (\mathbf{X^T X})^{-1}\mathbf{X^T}y$. Consequently, $\mathbf{P}y = \mathbf{X}\beta = \mathbf{X}^T(\mathbf{X^T X})^{-1}\mathbf{X^T}y$, for all $y \in \mathbf{R}^n$. This implies that

$$\mathbf{P} = \mathbf{X}(\mathbf{X^T X})^{\mathbf{-1}}\mathbf{X^T}.$$

Note that the inverse of the $k \times k$ matrix $\mathbf{X^T X}$ exists because $\rho(\mathbf{X}) = \rho(\mathbf{X^T X}) = k$.

2 Inverse of a matrix

2.1 Rank factorization of a matrix

Definition 5 *Let $\mathbf{A}$ be an $m \times n$ matrix with rank $r \geq 1$. Then $(\mathbf{P}, \mathbf{Q})$ is said to be a **rank factorization** of $\mathbf{A}$ if $\mathbf{P}$ is of the order $m \times r$, $\mathbf{Q}$ is of the order $r \times n$ and $\mathbf{A} = \mathbf{PQ}$.*

Let $\mathbf{A} = \mathbf{PQ}$ where $\mathbf{P}$ is an $m \times k$ matrix and $\mathbf{Q}$ a $k \times n$ matrix. Then $k \geq \rho(\mathbf{A})$. Further, the following are equivalent.

1. $k = \rho(\mathbf{A})$, i.e., $(\mathbf{P}, \mathbf{Q})$ is a rank factorization of $\mathbf{A}$.
2. $\rho(\mathbf{P}) = \rho(\mathbf{Q}) = k$, i.e., $\mathbf{P}$ is of full column rank and $\mathbf{Q}$ is of full row rank.
3. The columns of $\mathbf{P}$ form a basis of $\mathcal{C}(\mathbf{A})$.
4. The rows of $\mathbf{Q}$ form a basis of $\mathcal{R}(\mathbf{A})$.

Since $\mathbf{P}$ has a full column rank and $\mathbf{Q}$ has a full row rank, there are matrices $\mathbf{C}$ and $\mathbf{D}$ such that $\mathbf{CP} = \mathbf{I}$ and $\mathbf{QD} = \mathbf{I}$. The matrices $\mathbf{C}$ and $\mathbf{D}$ are called the *left inverse* of $\mathbf{P}$ and the *right inverse* of $\mathbf{Q}$, respectively.

2.2 Solution of linear equations

Given an $m \times n$ matrix $\mathbf{A}$, consider the set of linear equations $\mathbf{A}x = \mathbf{b}$, where $b \in \mathbb{R}^m$. This system of equations is said to be **consistent** if there exists a solution $x \in \mathbb{R}^n$ for this system of equations.

Proposition 6 *The following statements are equivalent.*

1. *$\mathbf{A}x = \mathbf{b}$ is consistent.*
2. *$b \in \mathcal{C}(\mathbf{A})$,*
3. *$\rho(\mathbf{A} : \mathbf{b}) = \rho(\mathbf{A})$.*

Proposition 7 *The system $\mathbf{A}x = \mathbf{b}$ is consistent iff $u^T\mathbf{A} = \mathbf{0} \Rightarrow u^T\mathbf{b} = 0$.*

Proof: If $\mathbf{Ay} = \mathbf{b}$ and $u^T\mathbf{A} = 0$, then $\mathbf{u^T b} = \mathbf{u^T Ay} = 0$. For the converse, note that the assertion states that $\mathbf{A}^T u = 0$ implies $\mathbf{b}^T u = 0$. Thus $u \in \mathcal{N}(\mathbf{A}^T)$ implies $u \in \mathcal{N}(\mathbf{b}^T)$. So $\mathcal{N}(\mathbf{A}^T) \subseteq \mathcal{N}\left(\begin{bmatrix}\mathbf{A}^T \\ \mathbf{b}^T\end{bmatrix}\right)$. The reverse inclusion trivially holds, so $\nu(\mathbf{A}^T) = \nu\left(\begin{bmatrix}\mathbf{A}^T \\ \mathbf{b}^T\end{bmatrix}\right)$. Note that, both the matrices has same number of columns. So they have the same rank. This implies $\rho(\mathbf{A}) = \rho(\mathbf{A} : \mathbf{b})$. So $\mathbf{A}x = \mathbf{b}$ is consistent.

Proposition 8 *Let S be the set of all solutions of a consistent system $\mathbf{A}x = \mathbf{b}$ and let u be any particular solution. Then $S = u + \mathcal{N}(\mathbf{A}) = \{u + w \ : \ w \in \mathcal{N}(\mathbf{A})\}$.*

Proof: Suppose $v \in S$. Then $\mathbf{A}v = \mathbf{b}$ and $\mathbf{A}u = \mathbf{b}$ implies $\mathbf{A}(v - u) = 0$. $v - u \in \mathcal{N}(\mathbf{A})$ and $v = u + (v - u) \in u + \mathcal{N}(\mathbf{A})$. Conversely, suppose $v \in u + \mathcal{N}(\mathbf{A})$. Then $v = u + w$ for some $u \in \mathcal{N}(\mathbf{A})$. So $\mathbf{A}v = \mathbf{A}u + \mathbf{A}w = \mathbf{b} + 0$ and $v \in S$.

Proposition 9 *If $\mathbf{A}$ has n columns then the system $\mathbf{A}x = b$ has*

1. *no solution if and only if $\rho(\mathbf{A}) < \rho(\mathbf{A} : \mathbf{b})$;*
2. *a unique solution if and only if $\rho(\mathbf{A}) = \rho(\mathbf{A} : \mathbf{b}) = n$;*
3. *at least two solutions if and only if $\rho(\mathbf{A}) = \rho(\mathbf{A} : \mathbf{b}) < n$.*

2.3 Generalized inverse of a matrix

Definition 6 *Let $\mathbf{A}$ be a $m \times n$ matrix. A matrix $\mathbf{G}$ of order $n \times m$ is said to be a generalized inverse (or g-inverse) of $\mathbf{A}$ if $\mathbf{AGA} = \mathbf{A}$.*

- We denote a g-inverse $\mathbf{G}$ of $\mathbf{A}$ by $\mathbf{A}^{-}$.
- If $\mathbf{A}$ is non-singular, $\mathbf{A}^{-1}$ is the unique g-inverse of $\mathbf{A}$.

Proposition 10 *Every $m \times n$ $\mathbf{A}$ matrix has a g-inverse.*

Proof: If $\mathbf{A} = \mathbf{0}$, any $n \times m$ matrix is a g-inverse of $\mathbf{A}$. So, let $\mathbf{A} \neq \mathbf{0}$. Then $\mathbf{A}$ has a rank factorization $(\mathbf{P}, \mathbf{Q})$. Let $\mathbf{CP} = \mathbf{I}$ and $\mathbf{QD} = \mathbf{I}$. Let $\mathbf{G} = \mathbf{DC}$. Then

$$\mathbf{AGA} = \mathbf{PQDCPQ} = \mathbf{PQ} = \mathbf{A},$$

which shows that G is a g-inverse of $\mathbf{A}$.

Proposition 11 *Let $\mathbf{A}, \mathbf{G}$ be matrices of order $m \times n$ and $n \times m$ respectively. Then the following are equivalent:*

1. *For any $y \in \mathcal{C}(\mathbf{A})$, $x = \mathbf{Gy}$ is a solution of $\mathbf{Ax} = \mathbf{y}$.*
2. *$\mathbf{G}$ is a g-inverse of $\mathbf{A}$.*
3. *$\mathbf{AG}$ is idempotent and $\rho(\mathbf{AG}) = \rho(\mathbf{A})$.*
4. *$\mathbf{GA}$ is idempotent and $\rho(\mathbf{GA}) = \rho(\mathbf{A})$.*

Proof:

(1) $\Rightarrow$ (2) Since $\mathbf{AGy} = \mathbf{y}$ for any $\mathbf{y} \in \mathcal{C}(\mathbf{A})$ for any $\mathbf{y} \in \mathcal{C}(\mathbf{A})$ we have $\mathbf{AGAz} = \mathbf{Az}$ for all $\mathbf{z}$.In particular, if $\mathbf{z}$ is the *ith* column of the identity matrix, then we see that the *ith* column of $\mathbf{AGA}$ and $\mathbf{A}$ are identical. So $\mathbf{AGA} = \mathbf{A}$.

(2) $\Rightarrow$ (1) For any $y \in \mathcal{C}(\mathbf{A})$, $y = \mathbf{Az}$ for some $\mathbf{z}$. Then $\mathbf{AGy} = \mathbf{AGAz} = \mathbf{Az} = \mathbf{y}$.

(2) $\Rightarrow$ (3) Clearly $\mathbf{AGAG} = \mathbf{AG}$. So $\mathbf{AG}$ is idempotent. Also

$$\rho(\mathbf{AG}) \leq \rho(\mathbf{A}) = \rho(\mathbf{AGA}) \leq \rho(\mathbf{AG}).$$

so that $\rho(\mathbf{A}) = \rho(\mathbf{AG})$.

(3) $\Rightarrow$ (2) $\rho(\mathbf{AG}) = \rho(\mathbf{A})$ implies $\mathbf{AGC} = \mathbf{A}$ for some $\mathbf{C}$. So $\mathbf{A} = \mathbf{AGC} = \mathbf{AGAGC} = \mathbf{AGA} = \mathbf{A}$. The second equality follows from $\mathbf{AG} = \mathbf{AGAG}$.

(2) $\equiv$ (4) Follows similarly as above.

Proposition 12 *If* $\mathbf{G}$ *is a g-inverse of* $\mathbf{A}$ *then* $\mathbf{AG}$ *is the projector into* $\mathcal{C}(\mathbf{A})$ *along* $\mathcal{N}(\mathbf{AG})$*.*

Proof : Since $\mathbf{AG}$ is idempotent, so $\mathbf{AG}$ is a projector into $\mathcal{C}(\mathbf{AG})$ along $\mathcal{N}(\mathbf{AG})$. Clearly $\mathcal{C}(\mathbf{AG}) = \mathcal{C}(\mathbf{A})$.

Here are a few useful facts about g-inverses.

Proposition 13 *1. If* $\rho(\mathbf{AB}) = \rho(A)$ *then* $\mathbf{B}(\mathbf{AB})^-$ *is a g-inverse of* $\mathbf{A}$*. If* $\rho(\mathbf{AB}) = \rho(B)$ *then* $(\mathbf{AB})^-\mathbf{A}$ *is a g-inverse of* $\mathbf{B}$*.*

2. If $\mathbf{A}$ *is a real matrix,* $\mathbf{A}^T(\mathbf{AA}^T)^-$ *and* $(\mathbf{A}^T\mathbf{A})^-\mathbf{A}^T$ *are g-inverses of* $\mathbf{A}$*.*

3. If $\mathcal{R}(\mathbf{A}) \subseteq \mathcal{R}(\mathbf{B})$ *and* $\mathcal{C}(\mathbf{C}) \subseteq \mathcal{C}(\mathbf{B})$ *then* $\mathbf{AB}^-\mathbf{C}$ *is invariant to the choices of* $\mathbf{B}^-$*.*

4. If $\mathbf{A}$ *is real and symmetric, so is* $\mathbf{A}^-$*.*

An important implication of the above, from the point of view of linear models, is that for any real matrix $\mathbf{X}$, $\mathbf{X}(\mathbf{X^TX})^-\mathbf{X^T}$ is symmetric and invariant to the choice of the g-inverse $(\mathbf{X}^T\mathbf{X})^-$.

2.4 Linear equations and g-inverses

The following is a characterization of solutions of linear equations in terms of g-inverses.

Proposition 14 *Let* $\mathbf{A}$ *be a* $m \times n$ *matrix, let* $y \in \mathcal{C}(\mathbf{A})$*. Then the class of solutions of* $\mathbf{A}x = \mathbf{y}$ *is given by* $\mathbf{A}^-\mathbf{y} + (\mathbf{I} - \mathbf{A}^-\mathbf{A})\mathbf{z}$ *for arbitrary* $\mathbf{z}$*.*

Definition 7 $\mathbf{A}^-$ *is called a* ***reflexive g-inverse*** *if* $\mathbf{A}^-\mathbf{AA}^- = \mathbf{A}^-$*.*

Observe that $\rho(\mathbf{A}) \leq \rho(\mathbf{A}^-)$. Furthermore, the equality holds if and only if $\mathbf{A}^-$ is reflexive.

Definition 8 *Suppose* $\mathbf{A}^-$ *satisfies* $(\mathbf{A}^-\mathbf{A})^T = (\mathbf{A}^-\mathbf{A})$*. Then for any* $y \in \mathcal{C}(\mathbf{A})$*,* $x = \mathbf{A}^-y$ *is a solution of* $\mathbf{A}x = y$ *with minimum norm. Such an* $\mathbf{A}^-$ *is said to be a* ***minimum norm g-inverse****.*

Definition 9 *Suppose* $\mathbf{A}^-$ *satisfies* $(\mathbf{AA}^-)^T = (\mathbf{AA}^-)$*. Then for any* x, y*,* $\|\mathbf{AA}^-\mathbf{y} - \mathbf{y}|| \leq ||\mathbf{Ax} - \mathbf{y}\|$*. Such* $\mathbf{A}^-$ *is a* ***least squares g-inverse*** *of* $\mathbf{A}$*.*

Definition 10 *If* $\mathbf{A}^-$ *is reflexive, minimum norm and least squares, then it is called a* ***Moore-Penrose inverse*** *of* $\mathbf{A}$*. It is usually denoted by* $\mathbf{A}^+$*.*

Proposition 15 *For any matrix* $\mathbf{A}$*, the Moore-Penrose inverse* $\mathbf{A}^+$ *exists and is unique.*

Proof: Start with the singular value decomposition of $\mathbf{A}$ as $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$, where $\mathbf{U}$ and $\mathbf{V}$ have orthonormal columns and $\mathbf{D}$ is a diagonal matrix with positive diagonal elements. Then, $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^T$. It is easy to verify that $\mathbf{A}^+$ satisfies all the conditions.

The following result provides a useful summary representation of the various objects encountered in this note.

Proposition 16 *Suppose the singular value decomposition of* $\mathbf{A}$ *be given by* $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^T$, *where* $\mathbf{U}$ *and* $\mathbf{V}$ *have orthonormal columns and* $\mathbf{D}$ *is a diagonal matrix with positive diagonal elements. Then the following holds:*

1. *The orthogonal projector into* $\mathcal{C}(\mathbf{A})$ *is* $\mathbf{P}_{\mathbf{A}} = \mathbf{U}\mathbf{U}^T$.
2. *The orthogonal projector into* $\mathcal{R}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^T)$ *is* $\mathbf{P}_{\mathbf{A}^T} = \mathbf{V}\mathbf{V}^T$.
3. *The Moore-Penrose inverse of* $\mathbf{A}$ *is* $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^T$.
4. $\mathbf{P}_{\mathbf{A}} = \mathbf{A}\mathbf{A}^+$.
5. *If* $\mathbf{A} = \mathbf{C}\mathbf{D}$ *is a rank factorization of* $\mathbf{A}$, *then the Moore-Penrose inverse*
$$\mathbf{A}^+ = \mathbf{D}^T(\mathbf{D}\mathbf{D}^T)^{-1}(\mathbf{C}^T\mathbf{C})^{-1}\mathbf{C}^T.$$
6. *If* $\mathbf{A}$ *is symmetric, then* $\mathbf{U} = \mathbf{V}$ *and* $\mathbf{A}$ *is a positive semidefinite matrix. Further, if* $\mathbf{U}$ *is a square matrix (hence, an orthogonal matrix), then* $\mathbf{A}$ *is positive definite.*
7. *If* $\mathbf{A}$ *is symmetric, then* $\mathbf{A}$ *is idempotent (and hence, a projection matrix) if and only if all the eigenvalues of* $\mathbf{A}$ *are either 0 or 1; if and only if* $\mathbf{D}$ *is the identity matrix.*
8. *If* $\mathbf{A}$ *is symmetric and idempotent, then* $tr(\mathbf{A}) = \rho(\mathbf{A})$.